Hydrothermal Helium Step Heats Reveal High Helium Retentivity and Multiple Diffusion Domain Systematics in Goethite

Ben J. Thyer[a,*], Jonathan Treffkorn[a], Kenneth A. Farley[a]

[a] *Division of Geological and Planetary Sciences, California Institute of Technology, Pasadena, 91125, CA, USA*

Abstract

U-Th/He geochronology of goethite (FeOOH) dates weathering and ferruginization directly, but interpreting a goethite date requires knowing how the mineral retains radiogenic helium, which has proven difficult to measure. Recent work revealed that goethite decomposes to hematite and water during vacuum step heating, and helium release records dehydration kinetics rather than helium migration through stable goethite. Goethite heated in vacuum develops pervasive contraction cracks visible with SEM, but this evidence of dehydration is absent in goethite heated at the same temperature under hydrothermal conditions. Thus, goethite step heats under hydrothermal conditions are expected to be less influenced by experimentally induced dehydration, and more likely to record helium loss in nature. We report the first helium step-heat experiment on goethite under hydrothermal conditions, made possible by sealing irradiated goethite and liquid water inside a pure silica glass ampoule, which retains water at up to 85 bar while transmitting helium to the vacuum system. Permeation through the ampoule wall delays measurement of helium released by the sample. We quantify this lag with

*Corresponding author.
Email address: bthyer@caltech.edu (Ben J. Thyer)

a first-order permeation model and remove it either by measuring the ampoule's residual helium with low-temperature permeation steps or, for long steps where the effect is comparatively minor, by bounding it by assuming the helium flux through the ampoule and sample are at steady state. Under this correction, encapsulated Durango apatite reproduces its published Arrhenius array, establishing that the ampoule's contribution to the signal can be separated from the sample's.

Proton-irradiated natural goethite sample Win-06-01, previously step heated in vacuum, was held in water vapor saturated conditions over 170–250°C for 40 days through multiple prograde and retrograde step heat cycles. Apparent helium diffusivities are approximately seven natural log units below those measured on the same sample in vacuum and its delta plot follows multiple diffusion domain systematics. Above 7% cumulative ^{3}He yield the array is linear, with an activation energy of 157.1 ± 5.3 kJ/mol. A domain with these parameters is nearly quantitatively retentive of helium at 25°C over billion-year timescales. Reanalysis of the vacuum step heats shows that they track first-order conversion of goethite to hematite. A nucleation model calibrated on them predicts that solid-state conversion at water saturation is slower by tens of orders of magnitude, and the hydrothermal release instead shows the history and time dependence of diffusion. We consider hydrothermal step heats to be a more faithful measure of helium migration in goethite than vacuum step heats, which are known to be compromised by dehydration. Whether these results are free of subtler alteration, and whether laboratory hydrothermal helium loss shares the same geometry as loss in nature, remain open questions a single experiment cannot resolve. The method developed

herein enables systematic tests of crystallite size, chemical composition, and radiation damage that could address them.

Keywords: goethite, U-Th/He geochronology, helium diffusion, hydrothermal, silica glass permeation, step heating

1. Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster et al., 2005; Farley et al., 2024). Goethite is a powerful chronometer for dating surface processes such as weathering and ferruginization directly: laterite and duricrust formation and recrystallization (Monteiro et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. Step heats of proton-irradiated, crystalline samples suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017), but the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved because goethite (FeOOH) decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley et al., 2024). Farley et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal,

governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Goethite dehydration is suppressed at high water pressure (Fig. 1), so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. A particularly simple and agile approach to do this is to seal the sample inside a pure silica glass ampoule. Silica glass is the most helium-permeable common glass (Norton, 1953; Altemose, 1961) because it lacks the network-modifying cations that occupy interstitial space in other glasses. Helium permeation in silica glass is sufficiently rapid, so helium evolved from a sample inside the ampoule escapes to the vacuum system and can be measured while other gaseous species, such as water, sulfur, or oxygen are retained. Earlier work has exploited the permeability and selectivity of silica glass for U-Th/He geochronology (Yakubovich et al., 2020, 2021, 2024) and for monotonic ramped-heating experiments (Yakubovich et al., 2019), but not to impose hydrothermal conditions during step heating.

Silica glass encapsulation offers a unique opportunity to preserve reactive phases such as goethite that would otherwise decompose during helium step heat experiments (Fig. 1), motivating the characterization of a step heat method utilizing this approach. Early applications reveal one cost: Permeation through the ampoule wall is rapid but not instantaneous, so helium released during a heating step is measured partly during that step and partly during later ones. We refer to this offset as permeation lag, and it must be quantified before any encapsulated step heat can be faithfully interpreted.

This paper establishes that the ampoule contribution to the measured he-

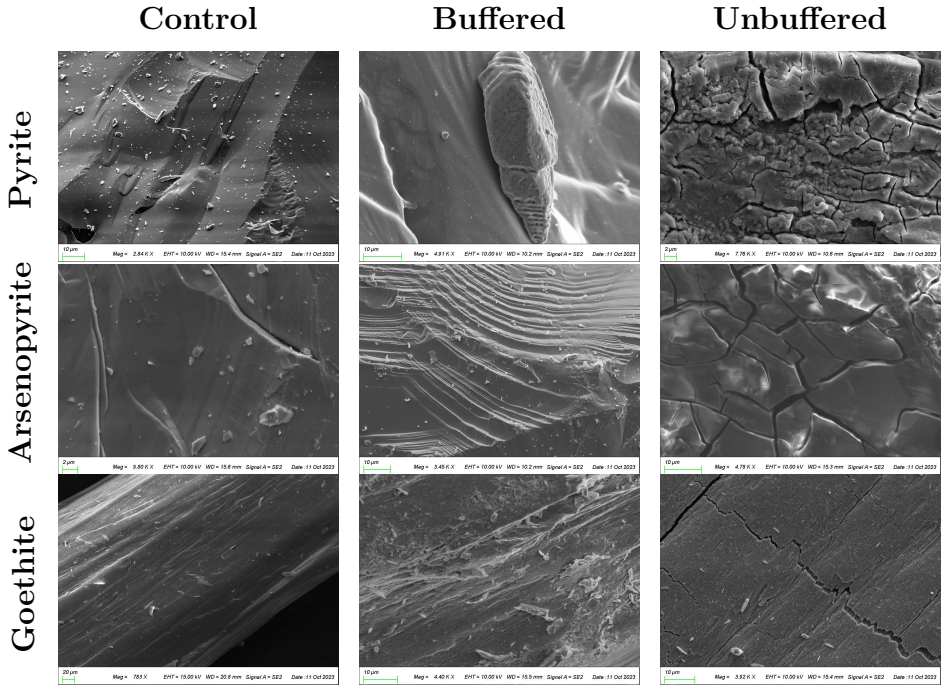


Figure 1: Secondary electron images of pyrite, arsenopyrite, and goethite before heat treatment (control), and after heat treatment with and without buffering in a silica glass ampoule with elemental sulfur, crystal realgar, and distilled water respectively. Unbuffered samples show contraction cracks and surface roughening; buffered samples retain the smooth surfaces and clear cleavage faces of the controls. For goethite, cracks in the unbuffered material run both parallel and perpendicular to the long axis of the polycrystalline aggregate.

lium in step heat experiments is tractable and that this method can address questions that vacuum experiments cannot. We show that encapsulation stabilizes reactive phases during heating and that we can successfully deconvolve the ampoule's permeation lag effect from helium step heat measurements of Durango apatite to recover its well-documented helium diffusion systematics (Wolf et al., 1998; Farley, 2000). We demonstrate that the ampoule's permeation rate constant can be measured during the experiment itself, which is necessary to apply the method to goethite because water-bearing ampoules often permeate helium anomalously fast. Finally, we apply the method to proton-irradiated goethite Win-06-01b under water saturation at 170–250°C, the first hydrothermal helium step heat reported for goethite.

2. Materials and Methods

2.1. Ampoule preparation and sealing

Segments of pure silica glass tube approximately 10 cm long, 4 mm outer diameter and 3 mm inner diameter, were sonicated in sequential baths of acetone, ethanol, and MilliQ water, then dried overnight at 100°C. The tubes were hand-pulled into pipettes with a jewelry welding torch and the thin ends were fused closed. The sample and buffering phase were then loaded into the pipette tip. Inner and outer diameters were measured on a calibrated stereo microscope. These give the geometric factor G of Eq. (7): 2390 cm^{-2} for the Durango apatite ampoule (850 µm inner, 1350 µm outer diameter) and 6980 cm^{-2} for the goethite ampoule (540 µm inner, 800 µm outer diameter).

It is paramount to avoid heating the sample during the encapsulation process to preserve the integrity of the step heat experiment; however, this is

challenging because silica glass must be heated well above 1600°C to fuse the ampoule. To protect against heating during the sealing process, the sample-end of Durango apatite ampoule was held by a water-soaked Kimwipe while the top was fused. The liquid water saturated the wipe in contact with the ampoules afterward, so the adjacent glass containing the sample never exceeded 100°C. For the goethite ampoule, the ampoule's contents were held in liquid nitrogen while the glass above was sealed closed, preventing the water and goethite from exceeding 0°C.

Before sealing, each ampoule was pumped to below 0.1 torr through its open top. The purpose of evacuation is mechanical: under an internal-external pressure contrast, softening glass deforms inward and self-seals, which gives a better seal with far less heat input. Although this process may ephemerally reduce the ampoule's atmospheric helium inventory, helium will be reintroduced fairly quickly: room-temperature permeation is fast enough that atmospheric helium re-enters a sealed ampoule within hours to days. Thus, it is advised to measure residual atmospheric helium directly at the start of each experiment (§2.7.3).

Ampoules containing water withstood the vapor pressure of liquid-water saturation up to about 86 bar at 300°C (Haar et al., 1984) but ruptured above 315°C. We recommend limiting water-bearing ampoules to below 300°C. Ampoules pulled from glass tubing of different sizes are likely to have different pressure tolerances.

2.2. Encapsulated Durango apatite

Durango apatite crushate sieved to 300 µm (radii ≈ 150 µm), 3.18 mg, was encapsulated (Fig. 2a) and subjected to step heats between up to 400

°C. Durango apatite is a standard for helium diffusion and U-Th/He thermochronometry (Zeitler et al., 1987; Wolf et al., 1996; Farley, 2000; Shuster and Farley, 2005; Shuster et al., 2006; Flowers et al., 2007), so its Arrhenius behavior is well-known independently. The experiment then is a direct test of the validity of the permeation-lag correction.

Each high-temperature step was preceded and followed by a sequence of two or three low-temperature permeation steps (§2.7.3). The total ^{4}He yield was estimated from the measured mass and the helium concentration of 194.2 nccSTP/mg reported by Farley (2000), a value verified independently in-lab. The atmospheric ^{4}He present at the start of the experiment, constrained by the initial permeation steps, was 1.47 ± 0.18 nccSTP. The full step-heat dataset is available in the data repository (Thyer, 2026).

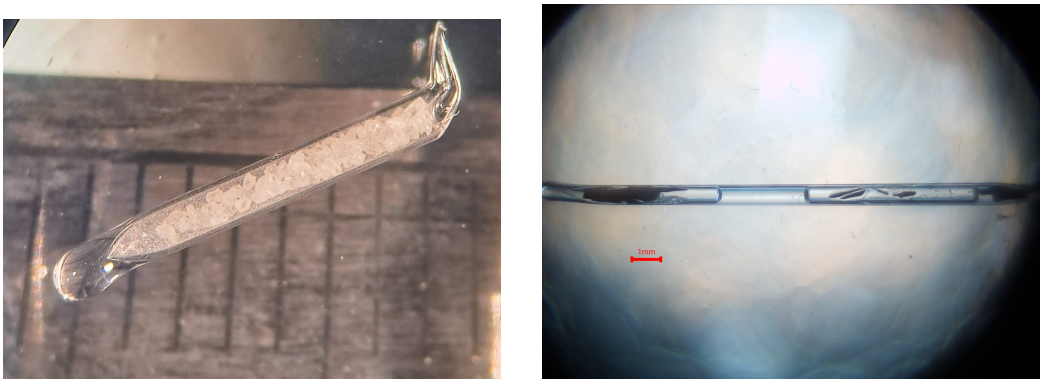
2.3. Hydrothermal goethite experiment

The goethite sample studied in this work is dense colloform goethite Win-06-01b, collected as float in the Flinders Ranges, South Australia (Waltenberg, 2012; Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages, 180–135 Ma (Waltenberg, 2012). Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following Shuster and Farley (2005) to produce a uniform ^{3}He distribution. Aliquots of this same irradiated material were previously step heated in vacuum by Monteiro (2017) and Farley et al. (2024), whose results agree well with one another.

An aliquot of 1.18 mg Win-06-01b goethite blades shorter than 2 mm was transferred to the ampoule tip in the presence of an ionizing fan. Less than one microliter of distilled water was added through a glass capillary, sub-

merging the goethite. The ampoule was tapped to release bubbles before it was flash-frozen in liquid nitrogen, then it was transferred to a −10°C saltwater-ice bath so that the water ice remained solid while atmospheric gases were pumped away. After pumping to <0.1 torr, the saltwater bath was swapped with liquid nitrogen to prevent heating of the sample or melting of the water ice buffer, and the ampoule was fused closed.

The sealed ampoule, designated HT-Win-06-01b, retained all of its original water through the entire experiment (Fig. 2b), across a pressure contrast of more than eleven orders of magnitude between an internal vapor pressure of roughly 40 bar at 250°C (Haar et al., 1984) and the high vacuum (10^{-7} torr) pressures of the lamp cell.



(a) (b)

Figure 2: Silica glass ampoules used for helium step heat experiments. (a) A silica glass ampoule containing Durango apatite crushate sieved to 300 µm (radii ≈ 150 µm). The ampoule is roughly one cm long and one mm wide; the bars visible behind it are separated by 1 mm. (b) HT-Win-06-01b after completion of the step heats. At room temperature this ampoule is about one third liquid water by volume. A gel-like lining on the inner wall is visible (see §4.7).

2.4. Heating and helium isotope analysis

Sealed ampoules were washed in ethanol and air dried, then placed in copper foil packets and mounted to a K-type thermocouple in an ultra-high-vacuum projector-lamp heating cell (Farley et al., 1999). For encapsulated Durango apatite, the lamp cell was pumped to high vacuum and held at 100°C for 30 minutes to desorb adsorbed species. Less than 0.5 ‰ of the radiogenic helium would be expected to escape during this pre-treatment, about 0.01 nccSTP, using the diffusion parameters of Farley (2000) and the forward equations of Fechtig and Kalbitzer (1966). The HT-Win-06-01b ampoule was instead held at 70°C for four days under vacuum to remove atmospheric helium entrained in the ampoule. The experiment then ran for about 40 days over 170–250°C, with step durations from 2 hours to about a week. We did not heat this sample above 250°C because of the risk of rupture due to high vapor pressure.

Helium isotope abundances for the Win-06-01b goethite were measured on a Thermo Fisher Helix-MC-Plus^{10K}, and the ^{4}He abundances for the encapsulated Durango apatite were measured with isotope dilution on a Pfeiffer QMS 200 mass spectrometer, both at Caltech. Evolved gases were purified over a liquid-nitrogen charcoal trap, a titanium sponge getter, and a SAES NP10 getter. For the Win-06-01b step heats, ^{3}He was counted on a pulse-counting electron multiplier and ^{4}He on a Faraday detector. Sensitivities were tracked with measurements of calibrated gas standards every 8 hours, giving internal reproducibilities of 5% for ^{3}He and 2% for ^{4}He (2σ).

For HT-Win-06-01b, a cold blank gave a negligible ^{3}He blank and a ^{4}He blank of about 1.25 pccSTP, comparable to the ^{4}He signals measured during

low-temperature permeation steps, but ^{3}He was well-above blank for these steps. Pump-out between steps was 1–5 minutes.

Temperature records during step heating show oscillations of about 1°C amplitude around each set point. While we did not conduct a detailed study to characterize temperature gradients throughout the ampoule during step heats, we suggest that the apatite experiment (§2.2, §3.1, §4.2) provides an independent check that recorded temperatures are accurate.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight at 50°C, then helium was completely degassed at ∼1100°C in a resistance furnace and measured on a Thermo Fisher Helix SFT mass spectrometer.

2.5. Electron microscopy of thermally treated phases

To assess whether encapsulation preserves reactive phases at experimental temperatures, single crystals of pyrite and arsenopyrite and aggregates of Bahia goethite morphologically similar to Win-06-01b (Shuster et al., 2005; Monteiro, 2017) were subjected to heat treatment in both buffered and unbuffered configurations: unbuffered samples were heated either in vacuum in the lamp-heating apparatus of Farley et al. (1999) or in open air, while buffered samples were sealed in silica glass ampoules with fused sulfur, mineral realgar, or water for pyrite, arsenopyrite, and goethite respectively. Sulfides were held at 400°C for at least 4 hours and goethite at 300°C for 4–11 hours. The ampoules were then broken and the samples air dried, mounted on carbon tape, carbon coated, and imaged in secondary electron mode on a ZEISS 1550VP field emission SEM at Caltech.

2.6. *Thermodynamic and kinetic analysis of goethite dehydroxylation*

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \;\rightleftharpoons\; \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}. \tag{1}$$

We evaluate its Gibbs energy change, $\Delta G_r(T, P)$, from the heat capacities, enthalpies and entropies of Majzlan et al. (2003b,a), refined against the phase-equilibrium brackets of Voigt and Will (1981), with water from the IAPWS-95 equation of state (Wagner and Pruß, 2002) as implemented in CoolProp (Bell et al., 2014). Grain-size effects use the hydrated surface energies of Navrotsky et al. (2008), comparing the two phases at equal surface area per mole of Fe. The driving force for hematite formation is $\Delta\mu \equiv -\Delta G_r$ per mole of Fe_2O_3. The formulation and input data are given in the Supplementary Material.

We reanalyze two published vacuum step heats of irradiated Win-06-01b. The experiment of Farley et al. (2024) (hereafter KAF) used 1 h steps from 70 to 330°C with the extraction line open to a liquid-nitrogen-cooled trap, except during two steps in which the trap was isolated, and the water released in each step was measured by capacitance manometry. The experiment of Monteiro (2017) (hereafter HDM) followed a cycled prograde–retrograde–prograde schedule to 498°C, and water could accumulate in its line. We describe conversion by a first-order rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A\exp\!\left(-\frac{E_a}{RT}\right), \tag{2}$$

where α is the fraction of goethite converted. For a powder of many grains, first-order kinetics is the limit in which each grain converts rapidly once a

hematite nucleus forms, so that k is a per-grain nucleation rate (Mampel, 1940). We assume that ^{3}He is released when its host grain converts, except for a fraction ϕ inherited by the hematite product, so that the cumulative fractional release is $F = (1 - \phi)\,\alpha$. Inverting Eq. (2) step by step gives an apparent rate constant for every step,

$$k_i = -\frac{1}{\Delta t_i} \ln \frac{1 - \alpha_i}{1 - \alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1 - \phi}, \tag{3}$$

which depends on no fitted parameter except ϕ. For comparison, the diffusion parameter D/a^2 of each step is obtained with the Fechtig and Kalbitzer (1966) inversion for a sphere. For HT-Win-06-01b, F and D/a^2 are taken from the bounding-method reduction (§2.7.5): cumulative release is bounded by the measured cumulative ^{3}He and by that value plus the upper-bound ampoule residual, central values are their midpoints, and steps without defined bounds or with near-blank yields are excluded, as in §3.3. Fitting windows and further details are given in the Supplementary Material.

2.7. Quantifying and Removing Permeation Lag

2.7.1. Helium budget of a single step

Consider one heating step. Helium enters the ampoule interior from the sample and leaves it by permeation through the wall, so

$$n_t = n_0 + \omega - m. \tag{4}$$

Here n_0 is the helium in the ampoule at the start of the step, n_t is the helium remaining at its end, ω is the helium released by the sample during the step, and m is the helium that permeated out and was measured. Only m is observed; but the quantity of interest is ω.

Steps are consecutive, so the helium inventory quantified between steps applies to both the previous and the subsequent step:

$$n_t^i = n_0^{i+1}, \qquad n_0^i = n_t^{i-1}. \tag{5}$$

Recovering ω for a step requires knowledge of the ampoule's helium content immediately before and immediately after it. Sections 2.7.3 and 2.7.5 give the two routes we use to obtain those quantities.

When ω is large compared with n_0 and n_t, Eq. (4) reduces to $\omega \approx m$ and no correction is needed. That condition does not hold for hour-long steps at 100–400°C, such as those used on the Durango apatite, but it is approached for long steps used with retentive phases, such as Win-06-01b (§2.7.5).

2.7.2. Assumptions and the permeation rate law

Helium permeation through silica glass is a well-understood first-order process (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968). Two conditions must hold before permeation can be described by a first-order rate law.

1. The helium partial pressure outside the ampoule is negligible compared with that inside. The ampoule and diffusion cell volumes differ by a factor of roughly 10,000 (Farley et al., 1999), so this holds throughout.
2. The helium concentration profile across the glass wall is at diffusive steady state. This is established within 30–70 minutes at 80°C for the ampoules used here, and it is disturbed whenever the temperature changes, because the glass/gas partitioning of helium is temperature dependent. We therefore hold the ampoule at the permeation temper-

ature for at least one diffusive timescale $\tau = L^2/D$ after every temperature change before measuring, where L is the wall thickness and D the diffusivity of helium in silica glass (Swets et al., 1961).

Given these conditions, permeation is first order in the amount of helium present:

$$\frac{dn}{dt} = -GKnRT, \tag{6}$$

where n is the number of helium atoms, G is a geometric factor with units of cm^{-2}, K is the permeation constant for helium in fused silica, which follows an Arrhenius law $K(T) = K_0 e^{-E_a/RT}$ and has been measured repeatedly (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), R is the gas constant, and T is temperature in Kelvin. Equation (6) is the standard permeation equation (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), written in terms of atom counts rather than partial pressures via the ideal gas law and taking the external partial pressure as zero.

For a cylindrical wall with negligible permeation through the ends, the geometric factor is

$$G = \frac{2}{a^2 \ln(b/a)}, \tag{7}$$

with a the internal and b the external radius (Masaryk, 1968). There is no dependence on length. G varies with ampoule construction and must be determined for each experiment.

2.7.3. Recovering the residual helium with permeation steps

At temperatures low enough that the sample releases no helium, $\omega = 0$ and the ampoule's helium inventory changes only by permeation. The measured helium from such a step is then a measurement of the helium inventory of the ampoule. To this end, a useful expression for the fractional yield of an isothermal step is derived, first through integration of Eq. (6) by separation of variables, yielding $n_t = n_0 e^{-GKRTt}$, then by relating the measured value to the tube contents by Eq. (4) ($m(\omega = 0) = n_0 - n_t$).

$$ {}^{He}F = \frac{m}{n_0} = 1 - e^{-\lambda t}, \qquad \lambda = KGRT. \tag{8} $$

This equation is applied to recover the helium inventory at the beginning of the step.

$$ n_0 = \frac{m}{{}^{He}F}, \tag{9} $$

where ${}^{He}F$ is the fraction of the ampoule's helium inventory that permeates out during a step of duration t, and λ is the ampoule's permeation rate constant. Collecting K, G, R, and T into λ is useful because λ is the quantity that can be measured directly during an experiment (§2.7.4), whereas K and G cannot be separated without independent knowledge of the ampoule geometry.

We call such a low-temperature step a *permeation step*. A permeation step does double duty: through Eq. (5) it constrains both the helium remaining after the preceding high-temperature step and the helium inherited by the following one. Placing a permeation step after every high-temperature step therefore supplies n_t and n_0 throughout the experiment, and Eq. (4) returns

ω. This is the approach used to reduce the encapsulated Durango apatite step heat data (§3.1).

2.7.4. Measuring the ampoule's rate constant during the experiment

Applying Eq. (9) requires λ. Published permeabilities for anhydrous silica glass (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968) combined with a measured G give an estimate, but they should not be relied upon, especially for hydrothermal experiments: water entering the glass changes its permeability (Shelby, 1972; Tsuru et al., 2011), and water-bearing ampoules show anomalously high rate constants (§4.3). The rate constant should therefore be measured on the ampoule actually in use.

Because permeation remains first order regardless of the values of K and G, a series of consecutive isothermal permeation steps determines λ and n_0 together with a non-linear least-squares solve.

$$m_{i+1} = n_0 \left(1 - e^{-\lambda t_{i+1}}\right) - n_0 \left(1 - e^{-\lambda t_i}\right). \tag{10}$$

We fit individual step yields rather than cumulative yields, because cumulative fitting propagates the errors of early measurements into later measurements, unduly biasing the solver to fit earlier measurements. We used the Levenberg–Marquardt implementation in SciPy (Virtanen et al., 2020), weighted by each measurement's own uncertainty, with initial guesses $\lambda \approx 6 \times 10^{-6}\,\mathrm{s}^{-1}$ and $n_0 \approx m_i/(1 - e^{-\lambda t_i})$ evaluated at the first measurement of each series. Uncertainties come from the fit covariance matrices. The fit is poorly conditioned unless enough time has elapsed for the cumulative yield to show curvature, which in practice means letting the permeation rate slow noticeably across three or more consecutive measurements. The exercise can

technically be done with as few as two consecutive measurements, although the solver would be unable to estimate uncertainties on the fit parameters.

2.7.5. Bounding the residual helium for long steps

Permeation steps require a measurable signal at low temperature. When the sample is retentive and its yields are small, the low-temperature signal can fall to blank level and Eq. (9) cannot be applied. This is the situation for ^{4}He in the goethite experiment (§3.2). An alternative approach is available when steps are long compared with the permeation timescale of the ampoule, which is common for retentive phases.

During a sufficiently long step the ampoule approaches a steady state in which the helium outflux of permeation is balanced by release from the sample, so the residual inventory stops growing while measurable helium continues to accumulate. As time proceeds after this condition is met, the helium inventory becomes relatively small compared to the measured abundance and $\omega \approx m$ becomes an increasingly accurate approximation. Additionally, an upper bound constraint on the ampoule's helium inventory can be calculated from the high-temperature experimental measurement alone. The residual is then bounded above by

$$n^{ss} \leq \frac{m}{\lambda t - 1}, \tag{11}$$

which holds once λt is large enough for steady state to be assumed. When Eq. (11) is applied to a long step and to the step preceding it, Eq. (4), can be used to bracket ω, the true abundance of helium evolved from the sample:

$$m_1 - \frac{m_0}{(\lambda t)_0 - 1} \leq \omega_1 \leq m_1 \left(1 + \frac{1}{(\lambda t)_1 - 1}\right). \tag{12}$$

Equation (11) assumes the sample's release rate is constant at a given temperature. In practice the release rate declines with time for volume diffusion (Wolf et al., 1996; Reiners and Farley, 1999; Farley and McKeon, 2015; Farley, 2018; Farley et al., 2024) and for first-order processes (Yakubovich et al., 2010; Wilson and Bisson, 1973), which makes the true residual smaller than the bound and Eq. (11) correspondingly conservative. The exceptions are initially inhomogeneous helium distributions at low cumulative yields (Goldsmith et al., 2020; Ketcham et al., 2011; Shuster et al., 2004; Farley et al., 1996) and radiation damage annealing (Cherniak, 2019; Guenthner et al., 2013; Flowers et al., 2007, 2009; Shuster et al., 2006), which can cause helium release to accelerate. Additionally, equation (11) is conservative when the value of λ used does not exceed the true one, because a lower bound on λ yields an upper bound on the residual. Any λ that can be independently shown to underestimate the ampoule's true rate constant is therefore admissible. This bounding approach is an excellent fallback when small yields during permeation steps of retentive phases cannot be well-resolved.

3. Results

3.1. Encapsulated Durango apatite step heats

Without correction, apparent diffusivities from the encapsulated Durango apatite step heat do not define the expected linear array (Fig. 3). After applying the permeation-step correction of §2.7.3, the corrected diffusivities agree within uncertainty with the Arrhenius relationship reported by Farley

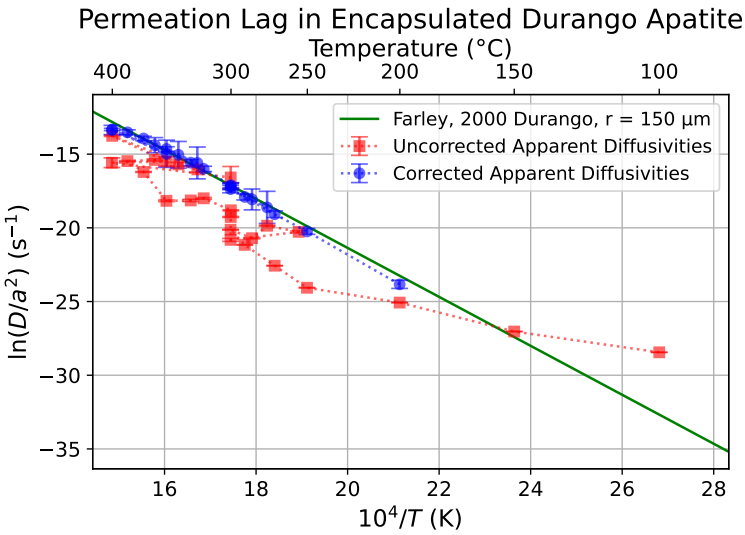


Figure 3: Arrhenius plot of a helium step heat of silica glass encapsulated Durango apatite with 2σ uncertainties, shown with and without the corrections of §2.7.3. The line corresponding to the diffusion parameters reported by Farley (2000) for the appropriate sieve size ($r = 150\,\mu m$) is shown as a reference. Apparent diffusivities are calculated with the equations of Fechtig and Kalbitzer (1966). Four corrected diffusivities are within 2σ of zero and are excluded.

(2000) for Durango apatite of the same 150 µm half-prism width, except for the first two plotted points and the 400°C step (Fig. 3), discussed below in §4.2. Four of the 25 high-temperature steps had permeation lag corrected yields within 2σ of zero and are excluded from the figure.

3.2. Helium permeation rate constants of the hydrothermal goethite ampoule

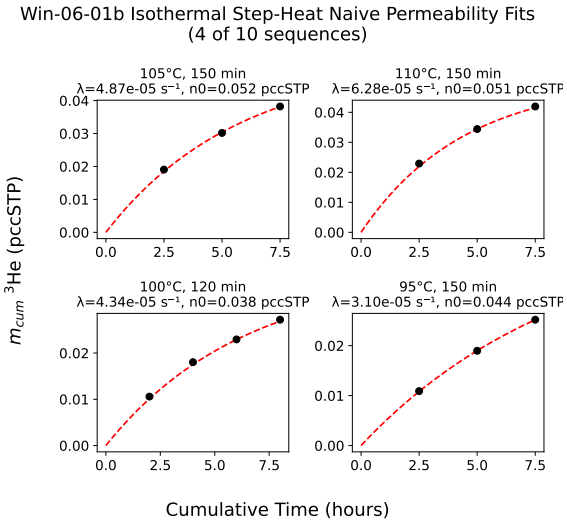


Figure 4: Cumulative ^{3}He yield against cumulative time for four of the ten series of consecutive isothermal permeation steps on HT-Win-06-01b, with the non-linear least-squares fit of λ and n_0 to Eq. (10) overlain, shown here as a representative example. Each series is held at a single temperature between 90 and 110°C. The rate constants recovered here are converted to permeabilities, $K = \lambda/GRT$, and plotted against temperature in Fig. 5.

^{3}He measurements from ten series of three to four consecutive isothermal permeation steps, held at 90–110°C for 120–180 minutes each with 26 minutes of cooling before inlet, were fit to Eq. (10) (Fig. 4). Fitted rate constants span 2.7×10^{-5} to $6.3 \times 10^{-5}\,\mathrm{s}^{-1}$, with 2σ uncertainties of 0.08 to 0.46 log

units and no trend resolvable against step duration. One further series that otherwise qualified was poorly conditioned and was excluded; however, its fitted λ agreed with the others within uncertainty. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly (§2.7.4).

Converting these rate constants to permeabilities through $K = \lambda/GRT$ and regressing $\ln K$ against inverse temperature over 90–110°C gives $R^2 = 0.936$ across 10 series, with an activation energy of 42.6 ± 7.9 kJ/mol (2 s.e.) (Fig. 5).

A finite-difference cooling simulation of permeation during the ampoule's cooling before measurement predicts that measurements will slightly overestimate the apparent permeability by an offset that is nearly constant across the temperature range. This offset is small: significantly smaller than the 2σ uncertainties on the fitted values (Fig. 5). The software to perform this simulation is included in the data repository (Thyer, 2026).

3.3. Helium release from goethite under hydrothermal conditions

The bounding method (§2.7.5) was used to reduce the HT-Win-06-01b step heat data (see §4.4 for details). The first three experimental steps have $\lambda t < 4$ and no bound could be evaluated; they are excluded from further analysis. Later steps with $\lambda t < 4$ could be calculated because bounds on the preceding steps' residuals were available. Six mid-experiment steps yielded near-blank yields. In total, 25 step heat measurements are plotted in Figs. 6, 7, 8, 9, and 12. The ampoule retained its full water charge for the duration of the experiment (Fig. 2b), so hydrothermal conditions were maintained throughout.

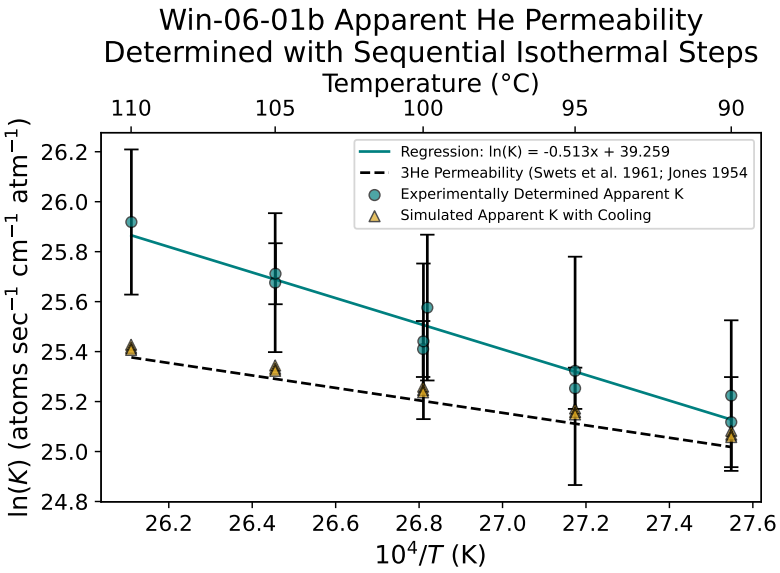


Figure 5: Apparent permeation coefficients for HT-Win-06-01b against inverse temperature, converted from the rate constants fitted in Fig. 4 via $K = \lambda/GRT$. Points labeled "Diffusion Model with cooling, Anhydrous" show the excess apparent permeability predicted by a numerical model for a dry ampoule that continued to permeate during the 26-minute cooldown before measurement; their scatter reflects differences in hold duration.

Apparent diffusivities for HT-Win-06-01b are lower than those measured for the same sample in vacuum by Monteiro (2017) and Farley et al. (2024) at every step. This gap widens to approximately seven natural log units after the first few steps (Fig. 6).

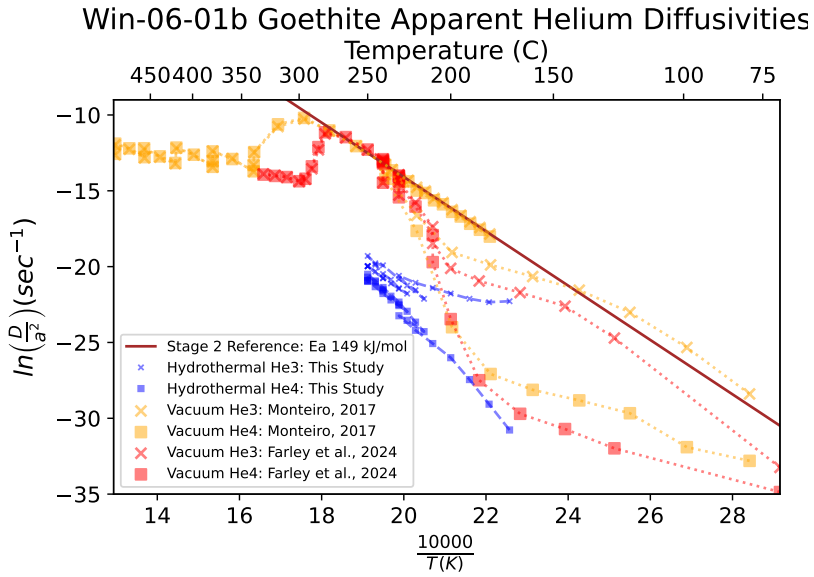


Figure 6: Arrhenius plot of ^{3}He and ^{4}He apparent diffusivities from HT-Win-06-01b, with step heats of the same Win-06-01b material in vacuum from Monteiro (2017) and Farley et al. (2024). The Stage 2 reference line described from Farley et al. (2024) is fit to the step heats of Monteiro (2017). Error bars on the data from this study are omitted for clarity.

The ^{3}He Arrhenius array sharply declines along a concave-up trend to about 3.5% cumulative ^{3}He yield, then decreases slowly along a sub-parallel linear array through the prograde and retrograde cycles between 3.5 and 7% cumulative yield (Figs. 7, 9). Beyond about 7% cumulative yield the array samples a well-defined line. A regression over the remaining 12 steps, span-

ning 7–15% cumulative ^{3}He yield, gives $R^2 = 0.989$ and an activation energy of 157.1 ± 5.3 kJ/mol. Treating this array as a single diffusion domain and considering only the uncertainty in the intercept gives a closure temperature of 98.1 ± 8.6 °C at a cooling rate of 10°C/Ma. This reference line is referred to as the "Stage 1" reference line, using the terminology of Farley et al. (2024) to refer to early helium losses in goethite preceding an increase in diffusivity. Total ^{3}He released over the experiment was 14.7% of the sample's total ^{3}helium inventory.

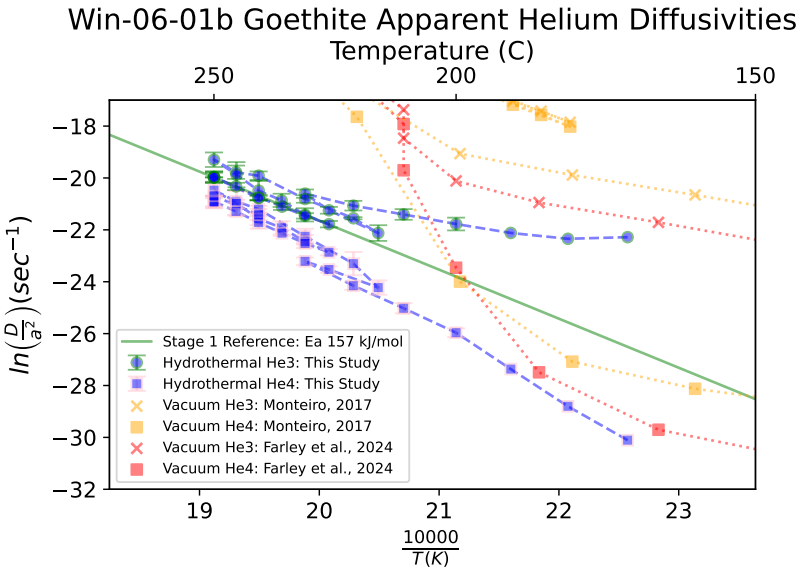


Figure 7: The linear regression of the ^{3}He apparent diffusivities of HT-Win-06-01b above 7% cumulative yield, defining the Stage 1 reference line. Apparent diffusivities calculated from ^{4}He generally lie about one log unit lower after a cumulative ^{3}He yield of 5%.

The ^{4}He array in Fig. 7 differs in shape. It is concave-down to about 5% cumulative ^{3}He yield, after which it follows a roughly linear array about one log unit below the ^{3}He array.

On a delta plot constructed against the ^{3}He regression as reference line, ^{3}He delta values decrease monotonically until about 7% cumulative loss and are within error of the reference line thereafter (Fig. 9). ^{4}He delta values gradually increase initially and settle into a slow increase or plateau about one log unit below the ^{3}He reference line by about 5% cumulative ^{3}He yield.

Bulk-normalized ^{4}He/^{3}He ratios begin at very low values and increase steeply, with the rate of increase slowing as the experiment proceeds (Fig. 12). At any given cumulative ^{3}He yield the hydrothermal experiment records lower bulk-normalized ratios than either vacuum experiment on the same sample, and its approach toward a plateau is the slowest of the three. The hydrothermal experiment did not reach a plateau.

4. Discussion

4.1. Encapsulation stabilizes reactive phases during heating

Pyrite, arsenopyrite, and goethite heated without a buffering phase, in vacuum or in open air, develop rough surfaces and contraction cracks (Fig. 1, right column), obvious signatures of chemical reaction. The same samples heated at the same temperatures inside sealed silica glass ampoules containing an appropriate buffer (sulfur, realgar, or water) retain the smooth surfaces and clear cleavage faces of the untreated controls (Fig. 1, center column). This observation shows that buffering some vacuum-unstable phases in silica glass ampoules exerts a strong stabilizing effect during step heat experiments, although this evidence does not preclude decomposition to a lesser extent that would not affect surface textures.

Step heat experiments that measure noble gas diffusivity in minerals under hydrothermal conditions are not new (Harrison, 1982; Harrison et al., 1985); however, the existing approach requires a new sample aliquot for each apparent diffusivity measurement and relies on precise knowledge of noble gas concentration a priori. The method introduced here for measuring helium migration has the distinct advantage that an arbitrary number of isothermal step heat measurements may be made for a single aliquot at a variety of temperatures, and the absolute concentration of helium for a given aliquot is directly measured. Furthermore, this method requires only limited alterations to well-established experimental protocols (Farley et al., 1999). Additionally, this methodological advancement is easily extended to sulfide phases that are unstable in vacuum (Fig. 1).

4.2. Recovering diffusivities for encapsulated Durango apatite

Step heats on encapsulated Durango apatite reproduce the diffusion behavior measured on unencapsulated material, following deconvolution of the permeation lag from the sample's own helium release using the protocol described in §2.7.3. The recovered inverted diffusivities of the encapsulated Durango apatite differ slightly from the reference line taken from Farley (2000) at the very beginning and end of the experiment, consistent with previous work on the sample. Farley (2000) also observed anomalously low apparent diffusivities in early steps. Above 300°C the corrected array curves downward, a feature documented for Durango apatite and attributed to annealing of radiation damage and crystal defects (Wolf et al., 1996; Farley, 2000; Shuster et al., 2006).

4.3. Elevated rate constants for helium permeation of water-bearing ampoules

Every ^{3}He permeability measurement for HT-Win-06-01b equals or exceeds the anhydrous literature value (Fig. 5), and the fitted activation energy of $42.6 \pm 7.9\,\mathrm{kJ/mol}$ is significantly higher than expected for silica glass with or without water (Swets et al., 1961; Jones, 1953; Masaryk, 1968; Shelby, 1972). Our measurement of a faster helium permeation rate for the HT-Win-06-01b ampoule than predicted by the anhydrous literature value for fused silica was, at first, an unexpected result. Water and hydroxyl groups dissolved in silica glass are reported to *decrease* helium permeability at all temperatures (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968), following the interpretation that species such as OH^- occupy interstitial space in the glass network (Norton, 1953; Shackelford, 1999); however, we have observed anomalously high rate constants in water-bearing ampoules consistently. The offset is real: it cannot be accounted for by the ampoule's extended permeation during cooling in the measurement procedure (§3.2).

The likely explanation is geometric: adding a significant volume of water reduces the volume available to the gas phase without reducing the glass area available for permeation. Helium is only sparingly soluble in water and diffuses through it rapidly, roughly $7 \times 10^{-5}\,\mathrm{cm^2/s}$ at room temperature (Verhallen et al., 1984) against about $2 \times 10^{-8}\,\mathrm{cm^2/s}$ in silica glass (Swets et al., 1961). On the timescale of permeation through the glass wall, transport through the water is effectively instantaneous, so the gas phase stays in equilibrium with the glass over the entire inner surface of the ampoule, including where liquid water contacts it. Helium partitions strongly into the gas phase over both water and glass: helium concentrations in the glass and water are

about 3% and 1%, respectively, of the gas-phase concentration, using solubilities from Sander (2015) and Swets et al. (1961). Displacing gas-phase volume with water therefore raises the helium partial pressure without changing the surface area available for permeation. Because the rate of helium permeation is proportional to the partial pressure in the gas phase (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), the permeability rate constant is related to the volume fractions of the gas and condensed phases within according to the following relationship, assuming that helium strongly partitions into the gas phase:

$$\lambda_{ampoule} = \lambda_{empty} \frac{V_{condensed} + V_{gas}}{V_{gas}}, \tag{13}$$

where $\lambda_{ampoule}$ is the rate constant of helium permeation for an ampoule containing a mineral/buffer payload, and λ_{empty} is the rate constant of that same ampoule with only the gas phase inside. This observed complexity emphasizes the need to measure the ampoule's rate constant in-situ during the experiment as described in §2.7.4.

4.4. Consistency of the bounding method to account for permeation lag for Win-06-01b

Both ^{3}He and ^{4}He data plotted in Figs. 6, 7, 8, 9, and 12 were reduced by the bounding procedure of §2.7.5 because ^{4}He could not be reduced following the preferred protocol of §2.7.3 due to blank-level permeation step yields. ^{3}He was reduced similarly for consistency with ^{4}He. As an internal test, the ^{3}He step heat datasets reduced using both approaches are intercompared here.

The bounding method produces the ^{3}He Stage 1 regression (§3.3) with activation energies between 157.1–160.6 kJ/mol and closure temperatures

between 97.5–101.2°C. Reducing ^{3}He instead by the direct route of §2.7.3 gives 157.0±4.8 kJ/mol (R^2=0.991, n=12, closure temperature 97.5±7.8°C), within uncertainty of the value obtained by bounding.

Fig. 8 shows this comparison directly: n_t, the ampoule's ^{3}He inventory quantified from permeation steps via Eq. (9), plotted against n^{ss}, the upper bound of Eq. (11) calculated independently using the preceding high-temperature step. Excepting two steps discussed below, $n_t \leq n^{ss}$ for every permeation step in the experiment, as required if the steady-state approximation genuinely bounds the ampoule's residual helium inventory. The agreement of the two ^{3}He datasets empirically validates the application of the bounding method.

The two very long "complete ampoule degas" permeation steps are in excess of the upper bound predicted by Eq. (11) (Fig. 8). These excesses are far greater than the expected diffusion from the Stage 1 reference domain of Fig. 7, which should be negligible at 100°C. This excess scales with step duration: the longer step (7200 minutes) exceeds n^{ss} by a factor of ≈1.5, whereas the shorter step (1590 minutes) exceeds n^{ss} by ≈25%. Because the relative excess seems to be roughly proportional to the step duration, it is unlikely that it is related to the first-order process of helium permeation through the ampoule, which would only produce a relative difference of a few percent between these two long steps. Thus, we hypothesize that this excess is caused by a diffusive fast pathway of the goethite sample (Lee and Aldama, 1992): in this case a slow bleed of helium from the interstices of the goethite aggregate during the low-temperature permeation step, recharged by helium liberated during the preceding high-temperature step. Although

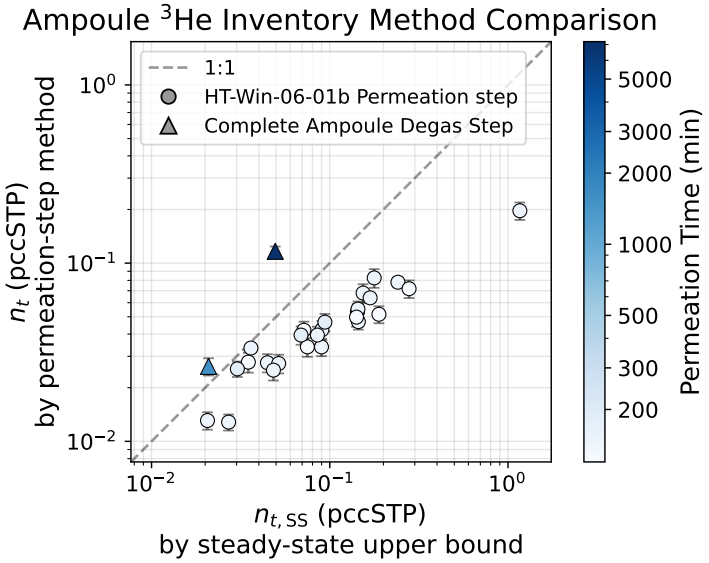


Figure 8: Two independent calculations of the ampoule's residual ^{3}He inventory following each high-temperature experimental step for HT-Win-06-01b. n_t (y-axis) is quantified directly from the low-temperature permeation step measurement via Eq. (9). n^{ss} (x-axis) is the upper bound of Eq. (11) calculated from the high-temperature experimental step. Color gives the permeation step duration. Circles correspond to n_t calculated from ordinary length permeation steps, whereas the triangles are two very long "complete ampoule degas" steps, held at 100°C for over a factor of ten longer than any other permeation step in the experiment. The dashed line is 1:1, representing the upper limit for self-consistent data.

there is insufficient data to make any interpretation confidently, it is unlikely that this effect would be caused by goethite dehydration, because the delta values of the high-temperature steps before and after these anomalous "complete ampoule degas" remain at zero (Fig. 9) and do not increase as would be expected for decomposition-controlled helium loss (Farley et al., 2024). Future studies may employ sequences of very long permeation steps following high-yield experimental steps to further study this attribute of helium migration in goethite.

4.5. Helium release from goethite is consistent with multiple diffusion domain systematics

Natural goethite occurs as aggregates of crystallites typically smaller than 1 µm, and all published goethite step heats are on polycrystalline material of unclear and probably variable crystallite size (Monteiro, 2017; Farley et al., 2024). Such a sample should show multiple diffusion domain (MDD) behavior, and neglecting it produces inaccurate activation energies and frequency factors (Farley, 2018).

Delta plots (Fig. 9) are a useful tool to diagnose MDD behavior: they remove the temperature dependence of the dominant migration process to show how apparent diffusivities evolve through the experiment. In a multiple diffusion domain system with an initially uniform diffusant (e.g. the initial ^{3}He distribution in HT-Win-06-01b), apparent bulk diffusivity at fixed temperature decreases as the smallest domains are exhausted, and delta values decrease correspondingly, provided the reference line's activation energy matches that of the controlling process (Lovera et al., 1991; Farley, 2018). Using the activation energy retrieved here, the ^{3}He delta values decrease

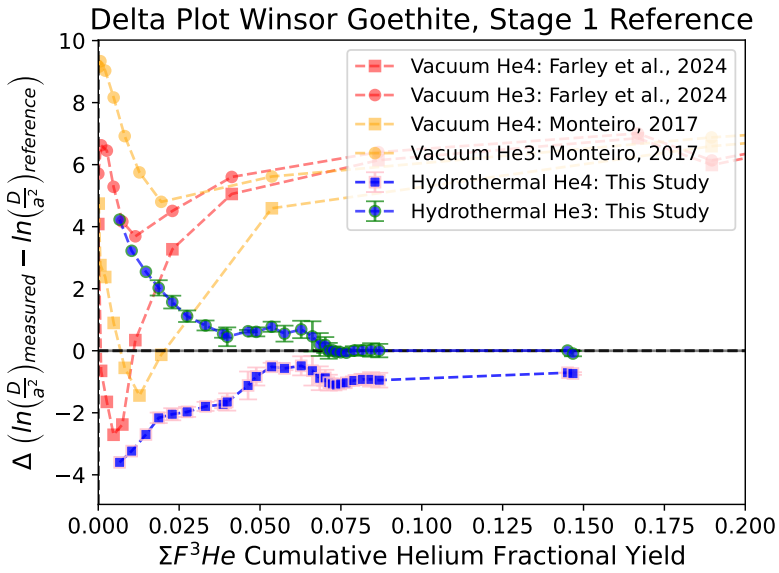


Figure 9: Delta plot after Farley (2018), comparable to the r/r_0 plots of Lovera et al. (1991) and Farley and Flowers (2012). Delta is the difference between apparent log diffusivity and the Stage 1 reference line (Fig. 7). ^{3}He decreases monotonically, then follows the reference line from 7–15%. ^{4}He delta values increase initially and settle about one log unit below the ^{3}He reference line.

monotonically (Fig. 9), which is the expected behavior.

The plateau in the delta plot and the quality of the regression over 7–15% cumulative ^{3}He yield ($R^2 = 0.989$) suggest that a single domain dominates the frequency factor over this interval, rather than a mixture of domains as is typically observed for polycrystalline hematite (Farley, 2018). The radiogenic ^{4}He apparent diffusivities are lower than the ^{3}He apparent diffusivities and increase rapidly early in the experiment when less retentive domains are degassed, which is an expected behavior for MDD systems that experienced some natural helium loss in nature (Shuster et al., 2004, 2005).

4.6. Hydrothermal helium release is diffusive, not dehydroxylation

Bulk goethite is metastable with respect to hematite and water at the conditions of every experiment considered here, in water or in vacuum; at 25°C bulk goethite requires $P_{H_2O} \approx 10^{2.6}$ bar to be stable (Supplementary Material). Stability alone does not determine what happens in a step heat, however. The rate of conversion depends on the driving force $\Delta\mu$, and because water is a product of Eq. (1), $\Delta\mu$ is set by the chemical potential of water. At water saturation, $\Delta\mu$ is only 6–11 kJ between 150 and 250°C, 1.6–2.4 times RT. Under the vacuum of a step-heating line ($P_{H_2O} \approx 10^{-8}$ atm) it is 76–106 kJ, about ten times larger.

The vacuum step heats of Win-06-01b behave as conversion to hematite. Step rate constants from both KAF and HDM (Eq. (3), $\phi = 0.112$) fall on a single Arrhenius line (Fig. 10a), including the HDM retrograde steps, as first-order kinetics requires. A least-squares fit to the 38 steps in the fitting window gives

$$E_a = 121.4 \pm 4.4 \text{ kJ mol}^{-1}, \qquad \ln A = 18.08 \pm 1.05, \tag{14}$$

with A in s^{-1}. In the diffusion representation the two experiments do not collapse, and HDM's retrograde steps plot about three natural-log units above its own prograde steps, which volume diffusion from progressively depleted domains cannot produce (Fig. 10b). Parameters fitted to the cumulative KAF release alone predict the independent HDM schedule with an rms misfit in F of 0.017, whereas a single-domain diffusion model misfits it by 0.053 and cannot produce the plateau at $F \approx 0.89$ reached by both experiments. That plateau is the fraction ϕ of ^{3}He inherited by hematite and released only by much slower diffusion from it; the conversion and the plateau correspond to the "Stage 2" and "Stage 3" release of Farley et al. (2024). The water evolved in each KAF step tracks the ^{3}He release, and in the two steps in which the trap was isolated and water accumulated, water and ^{3}He release slowed together, to 0.33 and 0.22 times the rate of neighboring steps, and recovered when the trap was restored. A diffusive process would not respond to the partial pressure of water.

The hydrothermal experiment behaves as diffusion. Under the vacuum kinetics of Eq. (14), its 442.8 h schedule would have converted the sample completely, yet only 14.7% of the ^{3}He was released. Two tests on the bounding-method reduction discriminate between the mechanisms (Fig. 11). First, the release depends on thermal history. The retrograde steps of the first temperature cycle at 225 and 215°C fall 0.45 and 0.86 natural-log units below the first prograde sweep in $\ln(D/a^2)$ (at least 0.18 and 0.47 at the bounds), and the 220°C step of the next prograde sweep sits 0.58 (at least 0.32) below its first-sweep value. In the last cycle, retrograde and prograde steps at 230–245°C agree within 0.07–0.45, with overlapping bounds. This is

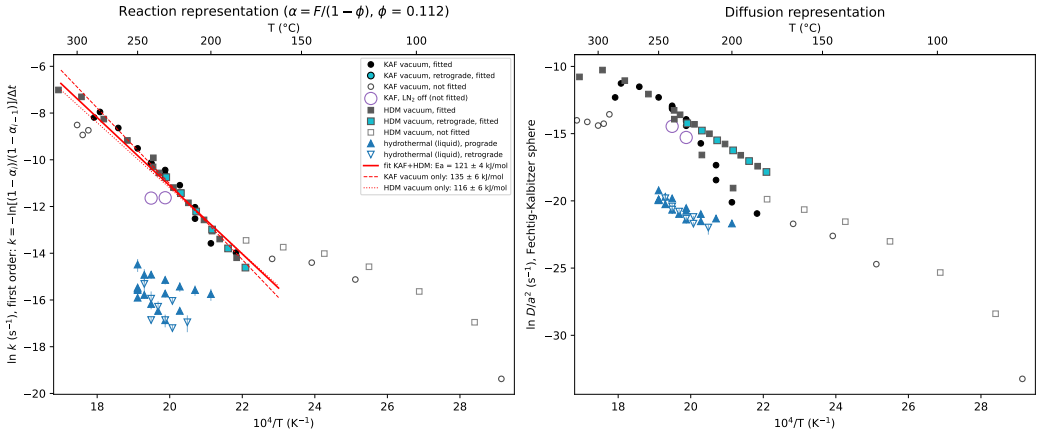


Figure 10: Every step of the two vacuum experiments on Win-06-01b (KAF, Farley et al. 2024; HDM, Monteiro 2017) and of HT-Win-06-01b in two representations. (a) First-order rate constants (Eq. (3), $\phi = 0.112$); red line: fit to KAF and HDM steps in the fitting window (filled symbols, Eq. (14)); open symbols: steps outside the window; purple circles: KAF steps with the LN_2 trap isolated. (b) Fechtig and Kalbitzer (1966) D/a^2. HT-Win-06-01b steps are bounding-method central values; bars span the lower and upper bounds. The vacuum experiments collapse in (a) but not in (b); the hydrothermal experiment shows the opposite behavior.

the signature of depleting fast, low-retentivity domains, whereas an interface-controlled conversion has almost no memory at $F \approx 0.04$. Second, the release slows with time at constant temperature. Calibrated on the three 5 h steps at 250°C, diffusion predicts $F = 0.148$ (0.139–0.158) at the end of the following 170.8 h step, bracketing the observed 0.1460–0.1463, whereas a constant-rate interface reaction overpredicts the ^{3}He released in that step by 39–60%. Plotted as first-order rate constants, the hydrothermal steps sit 1.3–2.8 orders of magnitude below the vacuum line and violate first-order behavior, with retrograde steps 0.7–1.5 natural-log units below prograde steps at the same temperature (Fig. 10a).

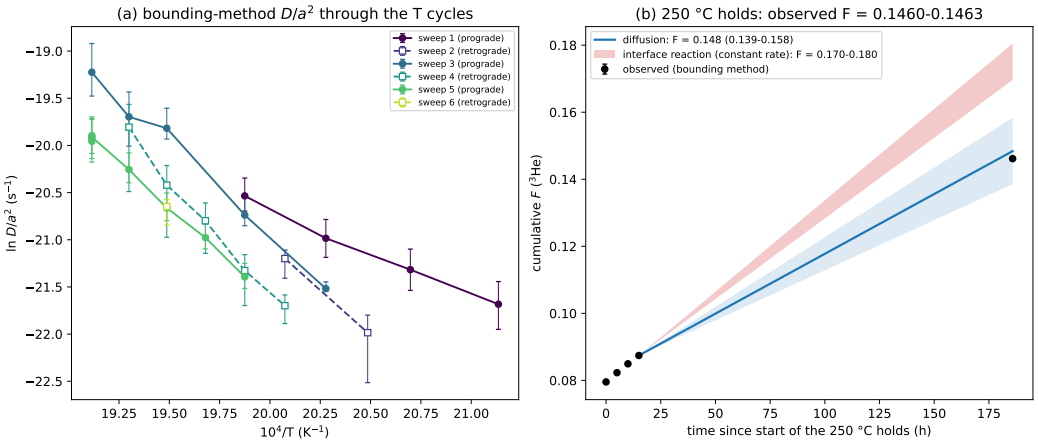


Figure 11: HT-Win-06-01b, bounding-method reduction (central values; bars span the lower and upper bounds; near-blank steps excluded). (a) Fechtig and Kalbitzer (1966) $\ln(D/a^2)$ through the temperature cycles (filled: prograde; open: retrograde). (b) The 250°C holds: the 170.8 h step compared with predictions calibrated on the three preceding 5 h steps for spherical diffusion and for a constant-rate, interface-controlled reaction; shaded bands span the predictions from the lower- and upper-bound reductions.

Water suppresses conversion because hematite formation is limited by

nucleation. In classical nucleation theory the barrier to forming a nucleus is

$$\Delta G^{*} = \frac{16\pi\,\gamma^{3}\,V_{\mathrm{hem}}^{2}}{3\,\Delta\mu^{2}}, \tag{15}$$

where γ is the energy of the interface surrounding the nucleus and V_{hem} the molar volume of hematite. The tenfold drop in $\Delta\mu$ between vacuum and water saturation raises the barrier about a hundredfold. The surface energies of Navrotsky et al. (2008) make nucleation on free surfaces far too slow to explain the vacuum kinetics, so hematite must nucleate internally, at a low-energy goethite–hematite interface. We calibrate the effective energy of that interface, $\gamma_{\mathrm{eff}} = 0.16 \pm 0.03\,\mathrm{J\,m^{-2}}$, from the slowdown of the two KAF steps with the trap isolated, bounding the water pressure in the line by the measured water and the apparatus volume. For the same sample, with the same interface and Arrhenius term, only $\Delta\mu$ changes between vacuum and saturation, and

$$\frac{k_{\mathrm{sat}}}{k_{\mathrm{vac}}} = \exp\left[-\frac{16\pi V_{\mathrm{hem}}^{2}\gamma_{\mathrm{eff}}^{3}}{3k_{B}T}\left(\frac{1}{\Delta\mu_{\mathrm{sat}}^{2}} - \frac{1}{\Delta\mu_{\mathrm{vac}}^{2}}\right)\right]. \tag{16}$$

Applied step by step to the hydrothermal schedule, Eq. (16) predicts a total conversion of $10^{-20.6}$–$10^{-53.7}$ for $\gamma_{\mathrm{eff}} = 0.14$–$0.19\,\mathrm{J\,m^{-2}}$, and the observed hydrothermal step rates exceed the predicted saturated conversion rates by 20 to more than 100 orders of magnitude. Far more ^{3}He is released in water than solid-state conversion could supply. A rate law linear in $\Delta\mu$ would predict only a 10–20% slowdown at saturation, so the suppression requires a barrier that grows as $\Delta\mu^{-2}$. This treatment rests on the assumptions that the trap-isolated slowdown acts through $\Delta\mu$ and that the interface energy and Arrhenius term are the same in vacuum and in water, and it describes

only solid-state conversion. The calibration, sensitivity tests and residuals are given in the Supplementary Material.

Fluid-mediated conversion is not described by Eq. (16), and it does occur in some materials. In qualitative experiments, acicular synthetic goethite pigment heated with excess water at 200–250°C for about a day converted either completely or not at all: roughly equal proportions of ampoules were fully red, fully yellow, or contained red and yellow portions isolated from one another by gas bubbles, and the reacted product consists of equant to platy crystals rather than pseudomorphs (Supplementary Material). This all-or-nothing outcome indicates a nucleation-limited dissolution–reprecipitation reaction that proceeds rapidly once it begins. Water saturation therefore suppresses solid-state conversion but does not by itself exclude fluid-mediated conversion, which depends on solution chemistry. For Win-06-01b in distilled water, the release is diffusive, and a nucleation-limited, all-or-nothing reaction would be expected to release helium abruptly rather than in the smooth, history-dependent manner observed.

4.7. Dehydration and recrystallization

Bulk goethite is unstable with respect to hematite and water at the pressures and temperatures of this experiment (Majzlan et al., 2003a), nanoparticles excepted (Guo and Barnard, 2011; Navrotsky et al., 2008). Solid-state dehydration is too slow at water saturation to contribute to the observed release (§4.6), but subtle dehydration, recrystallization and fluid-mediated conversion cannot be excluded on kinetic grounds alone. Electron microscopy reveals smooth and uncracked surface topography for water-buffered goethite, which excludes pervasive dehydration that would produce extensive cracks

in the sample, as in vacuum-heated samples (Fig. 1). This observation does not exclude dehydration below that scale, however, and work on the textural evolution of goethite aggregates at low dehydration fractions would be needed to resolve how much can occur before cracks appear.

Recrystallization is similarly difficult to exclude. Poorly crystalline synthetic goethite recrystallizes under hydrothermal conditions in as little as 16 hours at 125°C (Schwertmann et al., 1985), and our conditions were considerably harsher. The silica dissolved from the ampoule wall and redeposited as a gel-like lining (Fig. 2b) shows that the fluid was capable of dissolving and reprecipitating rock-forming oxides. Whether it recrystallized Win-06-01b specifically is unclear; this sample is among the most crystalline and most helium-retentive goethites yet dated (Waltenberg, 2012; Farley et al., 2024), whereas less crystalline goethites commonly recrystallize in nature and appear to lose their radiogenic helium (Monteiro et al., 2018, 2014; Heim et al., 2006).

The interpretive consequence is the same either way. If dehydration or recrystallization contributed to the release, the apparent diffusivities reported here are upper bounds on the true helium diffusivity in goethite at these temperatures and yields, and the retentivity we infer is a lower bound. This result would then imply that dehydration and recrystallization, rather than volume diffusion, set the pace of helium loss from goethite in nature.

4.8. Helium retentivity of goethite

Apparent diffusivities for HT-Win-06-01b are resolved from, and about seven natural log units below, those measured for the same sample in vacuum (Monteiro, 2017; Farley et al., 2024). We consider what this implies for

helium retentivity in nature through the lenses of closure temperature and isothermal retention. Both apply strictly only to loss by thermally activated volume diffusion from domains of known size, so each carries the caveats of §4.7, and none permits estimating the diffusivity at infinite temperature.

Treating the Stage 1 regression as a real diffusion domain gives a closure temperature of 98.1 ± 8.6 °C at 10°C/Ma, considering only the intercept uncertainty. However, closure temperature is an awkward measure for a mineral that forms at the surface and likely experienced limited burial, since it presumes a cooling rate (Dodson, 1973). Isothermal retention is generally more appropriate (Wolf et al., 1998). For a domain with the parameters of the Stage 1 reference held isothermally at 40°C for 100 Ma, $t'/t = 0.968$, where t' is the helium age, t the true holding time, and $t'/t = 1$ is perfect retention. At 25°C the same domain is essentially quantitatively retentive over billion-year timescales. Still, the ^{4}He apparent diffusivities are always lower than those for ^{3}He, suggesting that this relatively retentive domain corresponding to the Stage 1 reference experienced helium loss in nature.

4.9. Radiogenic helium loss recorded by $^4He/^3He$ ratios

Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples record the rounding of the radiogenic helium concentration profile by natural loss, provided loss in nature and in the laboratory propagate with the same geometry (Shuster et al., 2004; Farley et al., 2024). The steep initial rise and subsequent flattening seen in all three Win-06-01b experiments (Fig. 12) is consistent with complete, then partial, natural loss from the least retentive domains, perhaps from the interstices of the goethite aggregate (Shuster et al., 2005). The hydrothermal experiment likewise records low radiogenic

^{4}He retentivity at very low ^{3}He cumulative fractional yields, but it did not yield a plateau by the end of the experiment at a cumulative fractional ^{3}He yield of 15%.

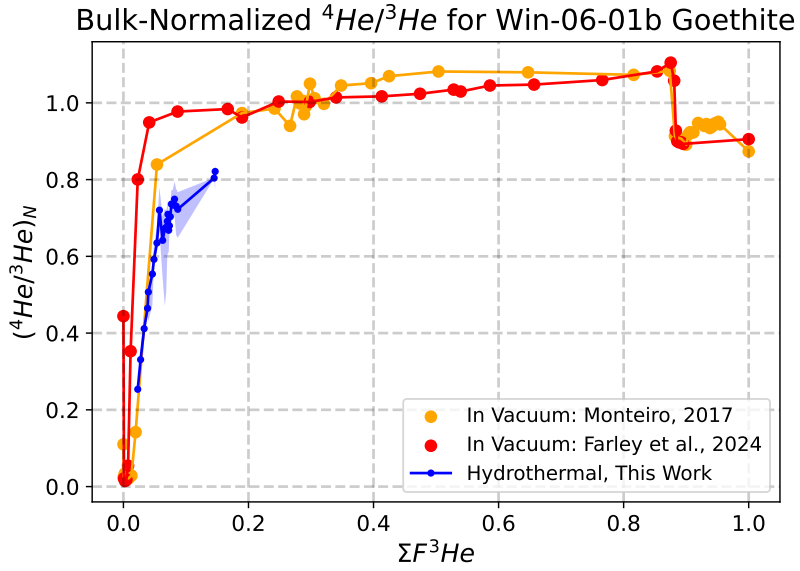


Figure 12: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b step heats against cumulative fractional ^{3}He yield. The hydrothermal experiment is less radiogenic than either vacuum experiment and is plotted as an envelope spanning the lower and upper bounds of Eq. (12).

Although the general trend is similar between the three experiments, the hydrothermal experiment records lower ratios for cumulative ^{3}He yields greater than ≈5% and approaches its plateau more slowly. This suggests that the geometry of helium loss differs between the hydrothermal and vacuum experiments. We interpret this experimental control on the ^{4}He/^{3}He spectra to be the relative importance of goethite dehydration and helium diffusion through crystalline goethite in governing the geometry of the helium loss from the sample, consistent with §4.6: in vacuum, grains release their inte-

grated inventory as they convert to hematite, whereas under hydrothermal conditions the release is diffusive. Thus, the geometry of helium loss due to dehydration and helium diffusion may differ, implying that the $^4He/^3He$ spectra of goethite measured in vacuum may not straightforwardly encode the geometry or history of helium loss in the same way that such data faithfully encodes this history in phases such as hematite or apatite (Farley and McKeon, 2015; Shuster et al., 2004). This claim entirely rests upon the correctness of the ^4He data reported here, for which each step yield is somewhat small, most often between 0.15 and 0.5 nccSTP. More $^4He/^3He$ spectra of goethite in hydrothermal conditions with stronger ^4He signals is needed to further investigate this hypothesis.

5. Conclusions

Sealing goethite in a silica glass ampoule with water enables helium step heat experiments under stabilizing hydrothermal conditions in a standard ultra-high-vacuum heating cell, sustaining internal water pressures up to about 86 bar at 300°C. Helium permeates the ampoule wall as a first-order process, and the lag this introduces between release and measurement can be removed either by measuring the ampoule's residual inventory with low-temperature permeation steps or, for long steps, by constraining it by assuming steady-state permeation. Using this framework, the expected Arrhenius array is recovered for encapsulated Durango apatite, demonstrating the tractability of the ampoule permeation correction. The ampoule's permeation rate constant can be measured during the experiment itself, which is necessary for hydrothermal work because water-bearing ampoules do not

follow published values for anhydrous glass.

The first hydrothermal helium step heat of goethite gives apparent diffusivities approximately seven natural log units below those obtained from the same sample in vacuum. The corrected ^{3}He array is linear between 7–15% cumulative yield, with an activation energy of 157.1 ± 5.3 kJ/mol, and its delta plot follows multiple diffusion domain systematics. Extrapolated to surface conditions, a domain with these parameters is essentially quantitatively retentive of helium at billion-year timescales at 25°C. The ^{4}He data require that this domain experienced helium loss in nature.

The ^{4}He/^{3}He spectra observed for this sample is distinct from the same sample in vacuum, suggesting that the geometry of helium loss induced by goethite dehydration is distinct from that of diffusion. More hydrothermal step heat experiments of goethite are necessary to confirm this hypothesis.

Hydrothermally treated goethite does not develop visible dehydration-induced contraction cracks like goethite heated in vacuum, and a thermodynamic and kinetic analysis of dehydroxylation explains why. Vacuum step heats of Win-06-01b follow first-order conversion to hematite, whereas at water saturation the roughly tenfold smaller driving force raises the barrier to hematite nucleation enough to slow solid-state conversion by tens of orders of magnitude; the hydrothermal release instead shows the history and time dependence of diffusion. Fluid-mediated conversion, which transformed synthetic goethite pigment in qualitative experiments, and recrystallization cannot be excluded on these grounds alone. If either contributed, the diffusivities reported here would be upper bounds and the retentivity a lower bound. Distinguishing these possibilities, and separating the effects of crys-

tallite size, chemical composition, and radiation damage, will require further hydrothermal experiments on irradiated single-crystal goethite of controlled size. The method described here makes such experiments straightforward.

Data Availability

Data are available through Zenodo at https://doi.org/10.5281/zenodo.22113166.

CRediT authorship contribution statement

Ben J. Thyer: Writing – Original Draft, Formal Analysis, Methodology, Investigation. **Jonathan Treffkorn:** Methodology, Investigation. **Kenneth A. Farley:** Writing – Review & Editing, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Science Foundation (NSF) grant EAR 1945974 to K.A. Farley. Dr. Paul Asimow is thanked for helpful discussions on the experimental design.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software programming, and with editing and LaTeX typesetting of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

Allard, T., Gautheron, C., Bressan Riffel, S., Balan, E., Soares, B.F., Pinna-Jamme, R., Derycke, A., Morin, G., Bueno, G.T., do Nascimento, N., 2018. Combined dating of goethites and kaolinites from ferruginous duricrusts: Deciphering the late neogene erosion history of central amazonia. Chemical Geology 479, 136–150. doi:10.1016/j.chemgeo.2018.01.004.

Altemose, V.O., 1961. Helium diffusion through glass. Journal of Applied Physics 32, 1309–1316.

Ansart, C., Quantin, C., Calmels, D., Allard, T., Roig, J.Y., Coueffe, R., Heller, B., Pinna-Jamme, R., Nouet, J., Reguer, S., Vantelon, D., Gautheron, C., 2022. (U-Th)/He Geochronology Constraints on Lateritic Duricrust Formation on the Guiana Shield. Frontiers in Earth Science 10. doi:10.3389/feart.2022.888993.

Bell, I.H., Wronski, J., Quoilin, S., Lemort, V., 2014. Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophys-

836 ical property library CoolProp. Industrial & Engineering Chemistry Re-
837 search 53, 2498–2508. doi:10.1021/ie4033999.

838 Cherniak, D.J., 2019. Diffusion of helium in radiation-damaged zircon.
839 Chemical Geology 529, 119308.

840 Danišík, M., Evans, N.J., Ramanaidou, E.R., McDonald, B.J., Mayers, C.,
841 McInnes, B.I.A., 2013. (u-th)/he chronology of the robe river channel iron
842 deposits, hamersley province, western australia. Chemical Geology 354,
843 150–162. doi:10.1016/j.chemgeo.2013.06.012.

844 Deng, X.D., Li, J.W., Shuster, D.L., 2017. Late mio-pliocene chemical weath-
845 ering of the yulong porphyry cu deposit in the eastern tibetan plateau
846 constrained by goethite (u-th)/he dating: Implication for asian summer
847 monsoon. Earth and Planetary Science Letters 472, 289–298.

848 Dodson, M.H., 1973. Closure temperature in cooling geochronological and
849 petrological systems. Contributions to Mineralogy and Petrology 40, 259–
850 274.

851 Farley, K.A., 2000. Helium diffusion from apatite: General behavior as il-
852 lustrated by durango fluorapatite. Journal of Geophysical Research: Solid
853 Earth 105, 2903–2914.

854 Farley, K.A., 2018. Helium diffusion parameters of hematite from a single-
855 diffusion-domain crystal. Geochimica et Cosmochimica Acta 231, 117–129.

856 Farley, K.A., Flowers, R.M., 2012. (U–Th)/Ne and multidomain (U–Th)/He
857 systematics of a hydrothermal hematite from eastern Grand Canyon. Earth
858 and Planetary Science Letters 359-360, 131–140.

Farley, K.A., McKeon, R., 2015. Radiometric dating and temperature history of banded iron formation-associated hematite, Gogebic iron range, Michigan, USA. Geology 43, 1083–1086.

Farley, K.A., Monteiro, H.B., Vasconcelos, P.M., Waltenberg, K., 2024. Dehydration of goethite during vacuum step-heating and implications for he retentivity characterization. Chemical Geology 663, 122254.

Farley, K.A., Reiners, P.W., Nenow, V., 1999. An apparatus for high-precision helium diffusion measurements from minerals. Analytical Chemistry 71, 2059–2061.

Farley, K.A., Wolf, R.A., Silver, L.T., 1996. The effects of long alpha-stopping distances on (u-th)/he ages. Geochimica et Cosmochimica Acta 60, 4223–4229.

Fechtig, H., Kalbitzer, S., 1966. The diffusion of argon in potassium-bearing solids, in: Schaeffer, O.A., Zähringer, J. (Eds.), Potassium Argon Dating. Springer, Berlin, Heidelberg, pp. 68–107.

Flowers, R.M., Ketcham, R.A., Shuster, D.L., Farley, K.A., 2009. Apatite (u-th)/he thermochronometry using a radiation damage accumulation and annealing model. Geochimica et Cosmochimica Acta 73, 2347–2365.

Flowers, R.M., Shuster, D.L., Wernicke, B.P., Farley, K.A., 2007. Radiation damage control on apatite (u-th)/he dates from the Grand Canyon region, Colorado Plateau. Geology 35, 447–450.

Goldsmith, A.S., Ketcham, R.A., Stockli, D.F., 2020. Simulating effects of heterogeneous 4he concentration profiles and radiation damage annealing

on whole-grain zircon diffusivity analyses. Geochimica et Cosmochimica Acta 284, 239–253.

Guenthner, W.R., Reiners, P.W., Ketcham, R.A., Nasdala, L., Giester, G., 2013. Helium diffusion in natural zircon: Radiation damage, anisotropy, and the interpretation of zircon (u-th)/he thermochronology. American Journal of Science 313, 145–198.

Guo, H., Barnard, A., 2011. Thermodynamic modelling of nanomorphologies of hematite and goethite. Journal of Materials Chemistry 21, 11566–11577.

Haar, L., Gallagher, J.S., Kell, G.S., 1984. NBS/NRC Steam Tables: Thermodynamic and Transport Properties and Computer Programs for Vapor and Liquid States of Water in SI Units.

Harrison, M.T., 1982. Diffusion of 40ar in hornblende. Contributions to Mineralogy and Petrology 78, 324–331.

Harrison, M.T., Duncan, I., McDougall, I., 1985. Diffusion of 40ar in biotite: Temperature, pressure and compositional effects. Geochimica et Cosmochimica Acta 49, 2461–2468.

Heim, J.A., Vasconcelos, P.M., Shuster, D.L., Farley, K.A., Broadbent, G., 2006. Dating paleochannel iron ore by (u-th)/he analysis of supergene goethite, Hamersley province, Australia. Geology 34, 173–176.

Hofmann, F., Reichenbacher, B., Farley, K.A., 2017. Evidence for >5 ma paleo-exposure of an eocene–miocene paleosol of the bohnerz formation, switzerland. Earth and Planetary Science Letters 465, 168–175.

Jones, W.M., 1953. Permeability and solubility of He^3 and He^4 in vitreous silica. Journal of the American Chemical Society 75, 3093–3096.

Ketcham, R.A., Gautheron, C., Tassan-Got, L., 2011. Accounting for long alpha-particle stopping distances in (u-th-sm)/he geochronology: Refinement of the baseline case. Geochimica et Cosmochimica Acta 75, 7779–7791.

Lee, J.K.W., Aldama, A.A., 1992. Multipath diffusion: A general numerical model. Computers & Geosciences 18. doi:10.1016/0098-3004(92)90093-7.

Lovera, O.M., Richter, F.M., Harrison, T.M., 1991. Diffusion domains determined by 39ar released during step heating. Journal of Geophysical Research: Solid Earth 96, 2057–2069.

Majzlan, J., Grevel, K.D., Navrotsky, A., 2003a. Thermodynamics of Fe oxides: Part II. enthalpies of formation and relative stability of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 855–859.

Majzlan, J., Lang, B.E., Stevens, R., Navrotsky, A., Woodfield, B.F., Boerio-Goates, J., 2003b. Thermodynamics of Fe oxides: Part I. entropy at standard temperature and pressure and heat capacity of goethite (α-FeOOH), lepidocrocite (γ-FeOOH), and maghemite (γ-Fe2O3). American Mineralogist 88, 846–854.

Mampel, K.L., 1940. Zeitumsatzformeln für heterogene Reaktionen an Phasengrenzen fester Körper. 2. Die Zeitumsatzformeln für ein Pulver aus

kugelförmigen Teilchen. Zeitschrift für Physikalische Chemie 187A, 235–249. doi:10.1515/zpch-1940-18727.

Masaryk, J.S., 1968. Permeation in fused silica. As cited in the fused-silica helium permeability literature (e.g., Shelby, 1972); original venue not confirmed.

Monteiro, H., 2017. Paleoenvironmental Evolution of Continental Landscapes through Combined High-Resolution Geochronology and δ^{18}O Ion Microprobe Analysis of Goethite. Ph.D. thesis. The University of Queensland. doi:10.14264/uql.2018.6.

Monteiro, H.S., Vasconcelos, P.M., Farley, K.A., Spier, C.A., Mello, C.L., 2014. (u-th)/he geochronology of goethite and the origin and evolution of cangas. Geochimica et Cosmochimica Acta 131, 267–289.

Monteiro, H.S., Vasconcelos, P.M.P., Farley, K.A., Lopes, C.A.M., 2018. Age and evolution of diachronous erosion surfaces in the amazon: Combining (u-th)/he and cosmogenic 3he records. Geochimica et Cosmochimica Acta 229, 162–183.

Navrotsky, A., Mazeina, L., Majzlan, J., 2008. Size-driven structural and thermodynamic complexity in iron oxides. Science 319, 1635–1638.

Norton, F.J., 1953. Helium diffusion through glass. Journal of the American Ceramic Society 36, 90–96.

Reiners, P.W., Farley, K.A., 1999. Helium diffusion and (u-th)/he thermochronometry of titanite. Geochimica et Cosmochimica Acta 63, 3845–3859.

Sander, R., 2015. Compilation of Henry's law constants (version 4.0) for water as solvent. Atmospheric Chemistry and Physics 15, 4399–4981.

Schwertmann, U., Cambier, P., Murad, E., 1985. Properties of goethites of varying crystallinity. Clays and Clay Minerals 33, 369–378.

Shackelford, J.F., 1999. Gas solubility in glasses – principles and structural implications. Journal of Non-Crystalline Solids 253, 231–241.

Shelby, J.E., 1972. Helium migration in natural and synthetic vitreous silica. Journal of the American Ceramic Society 55, 61–64.

Shuster, D.L., Farley, K.A., 2005. 4he/3he thermochronometry: Theory, practice, and potential complications. Reviews in Mineralogy and Geochemistry 58, 181–203.

Shuster, D.L., Farley, K.A., Sisterson, J.M., Burnett, D.S., 2004. Quantifying the diffusion kinetics and spatial distributions of radiogenic 4he in minerals containing proton-induced 3he. Earth and Planetary Science Letters 217, 19–32.

Shuster, D.L., Flowers, R.M., Farley, K.A., 2006. The influence of natural radiation damage on helium diffusion kinetics in apatite. Earth and Planetary Science Letters 249, 148–161.

Shuster, D.L., Vasconcelos, P.M., Heim, J.A., Farley, K.A., 2005. Weathering geochronology by (u-th)/he dating of goethite. Geochimica et Cosmochimica Acta 69, 659–673.

Swets, D.E., Lee, R.W., Frank, R.C., 1961. Diffusion coefficients of helium in fused quartz. The Journal of Chemical Physics 34, 17–22.

Thyer, B., 2026. Datasets for "hydrothermal helium step heats reveal high helium retentivity and multiple diffusion domain systematics in goethite". Zenodo. URL: https://doi.org/10.5281/zenodo.22113166, doi:10.5281/zenodo.22113166.

Tsuru, T., Igi, R., Kanezashi, M., Yoshioka, T., Fujisaki, S., Iwamoto, Y., 2011. Permeation properties of hydrogen and water vapor through porous silica membranes at high temperatures. AIChE Journal 57, 618–629. doi:10.1002/aic.12298.

Vasconcelos, P.M., Heim, J.A., Farley, K.A., Monteiro, H., Waltenberg, K., 2013. 40ar/39ar and (u-th)/he–4he/3he geochronology of landscape evolution and channel iron deposit genesis at Lynn Peak, Western Australia. Geochimica et Cosmochimica Acta 117, 283–312.

Verhallen, P.T.H.M., Oomen, L.J.P., Elsen, A.J.J.M.v.d., Kruger, J., Fortuin, J.M.H., 1984. The diffusion coefficients of helium, hydrogen, oxygen and nitrogen in water determined from the permeability of a stagnant liquid layer in the quasi-steady state. Chemical Engineering Science 39, 1535–1541.

Virtanen, P., et al., 2020. SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272.

Voigt, D.E., Will, G., 1981. TODO: verify title. Neues Jahrbuch für Mineralogie, Monatshefte 2, 89.

Wagner, W., Pruß, A., 2002. The IAPWS formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use. Journal of Physical and Chemical Reference Data 31, 387–535. doi:10.1063/1.1461829.

Waltenberg, K.M., 2012. Mineral physics and crystal chemistry of minerals suitable for weathering geochronology: Implications to 40ar/39ar and (u-th)/he geochronology. PhD thesis; institution not confirmed in the source draft.

Wilson, W.D., Bisson, C.L., 1973. Atomistics of helium diffusion in copper and tungsten. Radiation Effects 19, 53–58.

Wolf, R.A., Farley, K.A., Kass, D.M., 1998. Modeling of the temperature sensitivity of the apatite (u-th)/he thermochronometer. Chemical Geology 148, 105–114.

Wolf, R.A., Farley, K.A., Silver, L.T., 1996. Helium diffusion and low-temperature thermochronometry of apatite. Geochimica et Cosmochimica Acta 60, 4231–4240.

Yakubovich, O., Podolskaya, M., Vikentyev, I., Fokina, E., Kotov, A., 2020. U-th-he geochronology of pyrite from the Uzelga VMS deposit (South Urals) – new perspectives for direct dating of the ore-forming processes. Minerals 10, 629.

Yakubovich, O., Vikentyev, I., Ivanova, E., Podolskaya, M., Sobolev, I., Tyukova, E., Kotov, A., 2021. U-th-he geochronology of pyrite from alteration of the Au-Fe-Skarn Novogodnee-Monto deposit (Polar Urals,

Russia) – the next step in the development of a new approach for direct dating of ore-forming processes. Geosciences 11, 408. doi:10.3390/geosciences11100408.

Yakubovich, O.V., Gedz, A.M., Vikentyev, I.V., Kotov, A.B., Gorokhovskii, B.M., 2019. Migration of radiogenic helium in the crystal structure of sulfides and prospects of their isotopic dating. Petrology 27, 59–78.

Yakubovich, O.V., Konstantinova, N.P., Anosova, M.O., Podolskaya, M.M., Adamskaya, E.V., 2024. Technical note: An analytical approach for (U–Th) / He dating of goethite by sample encapsulation in quartz ampoules under vacuum, with an example from the Amerasian Basin, Arctic Ocean. Geochronology 6, 653–664. doi:10.5194/gchron-6-653-2024.

Yakubovich, O.V., Shukolyukov, Y.A., Kotov, A.B., Yakovleva, S.Z., Sal'nikova, E.B., 2010. Geothermochronology based on noble gases: II. stability of the (u-th)/he isotope system in zircon. Petrology 18, 555–570.

Yapp, C.J., Shuster, D.L., 2011. Environmental memory and a possible seasonal bias in the stable isotope composition of (u-th)/he-dated goethite from the canadian arctic. Geochimica et Cosmochimica Acta 75, 4194–4215.

Zeitler, P.K., Herczeg, A.L., McDougall, I., Honda, M., 1987. U-th-he dating of apatite: A potential thermochronometer. Geochimica et Cosmochimica Acta 51, 2865–2868.